

Orbit–Sliced Gram Energy at the Q^3 Gate: Gram–Copy Diagonal Closure, Collision Savings, and the Terminal Möbius Boundary

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July 2026

Abstract

We continue the distinguished zero–frequency analysis of a matched prime–Möbius cutoff shell. The preceding paper reduces the remaining physical amplitude to two column energies E_L, E_R and shows that

$$E_L + E_R \ll X^{o(1)} Q^3 \implies |\mathcal{S}_{\text{sh}}^{\text{all}}| \ll X^{o(1)} Q^2.$$

Here we perform the corresponding physical TT^* expansion. Opening the ultra divisor and the actual row-gcd projector gives an exact Gram kernel. If v_1, v_2 are the two projector divisors, then

$$|\mathcal{G}_L(\xi_1, \xi_2)| \ll X^{o(1)} \left(L + \frac{Q}{[v_1, v_2]} \right),$$

with a symmetric right-hand statement. This incidence gain degenerates on the principal layer $v_1 = v_2 = 1$, so it is not itself the required energy estimate.

We then isolate several unconditional pieces. The whole Gram-copy (same-input-row) diagonal satisfies

$$E_{L,\Delta} + E_{R,\Delta} \ll X^{o(1)} Q^2 J^2.$$

At $Q = X^{267/400+o(1)}$ and $J = X^{133/400+o(1)}$, this is $Q^3 X^{-1/400+o(1)}$: the complete Gram-copy diagonal crosses the Q^3 gate with a genuine power margin. The absolute target-collision mass with $m_1 j_1 = m_2 j_2$ is smaller, $O(X^{o(1)} Q^2 J)$. The absolute shared-ultra mass with $u_1 = u_2 > T \gg J$ is $O(X^{o(1)} Q^3 J^2/T)$, saving the full factor T from the absolute baseline but still missing the gate by J^2/T . On a fixed row it forces the two orbit times to be equal. More generally,

$$(mj + h, mj' + h) = (mj + h, j - j'),$$

which is the two-time counterpart of the earlier two-row target-content identity. The main reduction is an orbit-sliced energy. Put

$$Y_{\gamma,j}^L = \sum_{\alpha} \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad V_L = \sum_{\gamma,j} |Y_{\gamma,j}^L|^2,$$

and define V_R symmetrically. Exact Cauchy–Schwarz gives

$$E_L + E_R \ll X^{o(1)} J(V_L + V_R).$$

Consequently $V_L + V_R \ll X^{o(1)} Q^3/J$ is sufficient for the former Q^3 gate. Its orbit-sliced Gram-copy diagonal is only $O(X^{o(1)} Q^2 J)$, smaller than the allowance by $X^{1/400+o(1)}$. Thus

the unresolved object is a near-diagonal-scale, same-time averaged row correlation rather than the original two-time column energy.

Finally, on the high- β endpoint packet, the divisor range in each ultra increment contains the terminal value $u = mj + h$; its coefficient is zero when $\mu(mj + h) = 0$. On this layer the row and target signs fuse as $\mu(d)\mu(\ell dj + h)$, and TT^* produces an averaged four-Möbius row correlation. An exact coherent zero-pair-diagonal model attains $E \asymp Q^3 J^2$, proving that cardinality, equal-pair deletion, and absolute envelopes alone cannot yield the desired Q^3 bound. This is a method obstruction, not a lower bound for the physical energy. No affine Chowla estimate, prime-pair asymptotic, positivity theorem, twin-prime theorem, or breach of sieve parity is claimed.

Keywords: Möbius correlation; TT^* dispersion; Gram energy; orbit slicing; divisor collision; zero frequency; parity boundary.

MSC 2020: Primary 11N35, 11N37; Secondary 11B25, 11L07, 15A18.

1 Introduction

The zero-frequency coefficient left by the matched cutoff construction is no longer an unspecified parity remainder. TPC-32 identifies the exact distinguished coefficient and the $1/400$ endpoint ledger [9]; TPC-33 returns to the physical row mask, opens a primitive affine column, proves a fixed-determinant finite-field second moment, and reduces the scalar shell to the energy condition

$$E_L + E_R \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^3. \quad (1)$$

The implication from (1) to the required Q^2 scalar amplitude is proved in TPC-33 and is recalled in section 2. The estimate itself is not inherited [10].

The purpose of this paper is to determine what a literal TT^* analysis does and does not prove at that gate. This is useful for two reasons. First, the column energy hides two independent duplications: a row duplication and an orbit-time duplication. Treating both as one generic off-diagonal discards a small but decisive endpoint margin. Second, opening the actual row-gcd projector creates genuine divisor incidence, but the principal divisor $v = 1$ survives with no gain. The remaining obstruction can therefore be stated as a concrete averaged Möbius row energy rather than as a vague request for more pseudorandomness.

The algebraic TT^* principle and the large sieve are classical [1, 3]. The contribution here is not a new abstract Hilbert-space inequality. It is the exact placement of that inequality inside the physical matched shell, including the true mask, the ultra-divisor variable, the endpoint exponents, and the sign-sensitive terminal layer.

1.1 Main results

Let

$$K_{\gamma, \alpha}^L = \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_\alpha}(j) H_{m_\gamma}(j), \quad E_L = \|K^L \gamma^{(1)}\|_2^2,$$

with K^R, E_R defined by exchanging the two rows and the two cutoff legs. Opening both the ultra increment C and the row-gcd projector gives an exact synthesis matrix $T_L = (\Phi_\gamma(\xi))$, where $\xi = (\alpha, j, u, v)$, and

$$E_L = \sum_{\xi_1, \xi_2} \overline{z_L(\xi_1)} z_L(\xi_2) \mathcal{G}_L(\xi_1, \xi_2), \quad \mathcal{G}_L = T_L^* T_L. \quad (2)$$

The physical opposite-row count gives

$$|\mathcal{G}_L(\xi_1, \xi_2)| \ll X^\varepsilon \left(L + \frac{Q}{[v_1, v_2]} \right). \quad (3)$$

This is strongest when the two gcd-projector divisors have a large least common multiple and is trivial on $v_1 = v_2 = 1$.

The first positive theorem closes the entire Gram-copy (same-input-row) diagonal:

$$E_{L,\Delta} + E_{R,\Delta} \ll X^\varepsilon Q^2 J^2. \quad (4)$$

When $Q = X^\beta$, $J = X^{1-\beta}$, this crosses the Q^3 gate as soon as $\beta > 2/3$. At the physical endpoint $\beta = 267/400$, its ratio to Q^3 is exactly

$$\frac{Q^2 J^2}{Q^3} = \frac{J^2}{Q} = X^{-1/400+o(1)}. \quad (5)$$

The margin is small, but it is a real power and no conjectural input is used.

Two further collision estimates locate the remaining mass. The absolute Gram mass with equal targets, $m_{\alpha_1} j_1 = m_{\alpha_2} j_2$, is

$$\ll X^\varepsilon Q^2 J. \quad (6)$$

The absolute mass with the same ultra divisor $u_1 = u_2 > T \gg J$ is

$$\ll X^\varepsilon \frac{Q^3 J^2}{T}. \quad (7)$$

Thus shared-divisor incidence saves the factor T from the absolute baseline but still misses the Q^3 gate by J^2/T . On the same row, however, a shared ultra divisor forces the orbit times to coincide, because

$$(mj + h, mj' + h) = (mj + h, j - j'). \quad (8)$$

The cleanest next interface is obtained before duplicating orbit time. Define V_L, V_R by freezing j inside the row sum. Then

$$E_L + E_R \ll X^\varepsilon J(V_L + V_R). \quad (9)$$

Hence

$$V_L + V_R \ll X^\varepsilon Q^3/J \quad (10)$$

is sufficient for (1). The Gram-copy diagonal of this new energy is $O(X^\varepsilon Q^2 J)$, and

$$\frac{Q^3/J}{Q^2 J} = \frac{Q}{J^2} = X^{1/400+o(1)}. \quad (11)$$

The permissible total is therefore only a small power above the natural diagonal scale. Proving the stronger diagonal-scale estimate $V_L + V_R \ll X^\varepsilon Q^2 J$ would return

$$E_L + E_R \ll X^\varepsilon Q^2 J^2 \ll_{\mathscr{D}} X^{2+\varepsilon} = X^{\varepsilon+o(1)} Q^3 X^{-1/400}.$$

1.2 The terminal sign boundary

On the high- β endpoint packet, (38) gives $T < N_m(j) < U_0$. Thus the divisor range in $C_m(j)$ contains the terminal value $u = N_m(j) = mj + h$. Its coefficient is

$$a_R(N_m(j)) = -\mu(N_m(j)) \log N_m(j), \quad (12)$$

and vanishes when $\mu(N_m(j)) = 0$. Together with the row sign $\mu(d)$, this produces $\mu(d)\mu(ldj + h)$. The same-time energy consequently contains an average of four Möbius factors after expansion. Existing fixed-form or logarithmically averaged Chowla results do not directly provide the growing-coefficient, dyadic, physically masked estimate (10); see the precise boundary already recorded in TPC-33 and the primary sources cited there [4, 5, 10].

An abstract sign-blind model satisfying the same cardinality and pointwise envelopes has $E \asymp Q^3 J^2$. It follows that cardinality, equal-pair deletion, and pointwise envelopes alone cannot establish (1) uniformly over that abstract class. This statement is deliberately methodological. It is not a physical lower bound, and it leaves open a proof using the fused Möbius signs or cancellation between terminal and proper-divisor layers, as well as a proof using additional arithmetic incidence. The appearance of this sign-sensitive boundary is consistent with the classical parity problem in sieve theory [2].

1.3 Organization and claim boundary

Section 2 imports the physical definitions. Section 3 proves the exact Gram expansion and its divisor-incidence kernel bound. Section 4 closes the Gram-copy diagonal and target collision, proves the shared-ultra and two-time content estimates, and distinguishes savings from closure. Section 5 proves the orbit-sliced transfer. Section 6 isolates the terminal Möbius layer and the sign-blind obstruction. Section 7 gives the exact endpoint ledger and the next theorem menu. The accompanying certificate checks only finite algebra and rational bookkeeping.

No estimate of the off-diagonal energy in (10) is proved. In particular, this paper does not prove an unweighted affine Chowla theorem, the zero-frequency flatness statement of TPC-32, a Hardy–Littlewood prime-pair asymptotic, positivity, infinitely many twin primes, or a breach of sieve parity.

2 The inherited physical interface

We use the notation and proved bounds of TPC-32 and TPC-33 [9, 10]. They are restated here so that every new estimate can be checked without changing the physical packet.

2.1 Rows, orbit, and cutoff factors

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and work on one opened dyadic packet with

$$Q = LD, \quad QJ \asymp_{\mathcal{D}} X, \quad R \leq T < U_0 \asymp_{\mathcal{D}} X. \quad (13)$$

The number U_0 is chosen above every target. A row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp_{\mathcal{D}} Q, \quad (14)$$

where $\ell_\alpha \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, $(d_\alpha, H) = 1$, and m_α determines its row label. In particular,

$$\#\{\alpha\} \ll_{\mathcal{D}} Q. \quad (15)$$

The two coefficient systems have the form

$$\gamma_\alpha^{(i)} = \mu(d_\alpha)(\log \ell_\alpha)\omega_D^{(i)}(d_\alpha)\psi_L^{(i)}(\ell_\alpha/L)\zeta_\alpha^{(i)}, \quad i = 1, 2, \quad (16)$$

and satisfy

$$\sum_{\alpha} |\gamma_\alpha^{(i)}| \ll_{\mathcal{D}} Q, \quad (17)$$

$$\sum_{\alpha} |\gamma_\alpha^{(i)}|^2 \ll_{\mathcal{D}} Q \log(2L). \quad (18)$$

For j in a fixed interval containing the joint support, of length $O_{\mathcal{D}}(J)$, define

$$N_\alpha(j) = m_\alpha j + h. \quad (19)$$

Every effective term has $(m_\alpha j, H) = 1$, every target is $\asymp_{\mathcal{D}} X$, and each divisor $w \mid N_\alpha(j)$ on the physical support satisfies

$$(w, m_\alpha j H) = 1. \quad (20)$$

Since $H = \text{rad } |h|$, the first condition implies $(m_\alpha, h) = 1$; the displayed divisor condition also implies $(w, m_\alpha) = 1$. For sufficiently large X , the target condition together with $m_\alpha \asymp_{\mathcal{D}} Q$ implies $j \asymp_{\mathcal{D}} J > 0$.

Here $\kappa_0 > 0$ is a fixed pruning parameter. The actual row-pair mask and multiplier are

$$\mathbf{m}_{\text{act}}(\alpha, \gamma) := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > QX^{-\kappa_0}} \mathbf{1}_{(d_\alpha, d_\gamma) \leq G}, \quad G = X^{\kappa_0}, \quad (21)$$

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) := \mathbf{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (22)$$

The fixed-period and smooth factors are uniformly bounded. It will be convenient to remove only the row-gcd indicator and write

$$\mathfrak{B}_{\alpha, \gamma}(j) := \mathbf{1}_{\ell_\alpha \neq \ell_\gamma} \mathbf{1}_{|m_\alpha - m_\gamma| > QX^{-\kappa_0}} \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J). \quad (23)$$

2.2 Ultra increment and matched polarization

Let

$$a_R(u) = -\mu(u) \log u \quad (24)$$

and define

$$C_m(j) := \sum_{\substack{T < u \leq U_0 \\ u \mid mj+h}} a_R(u), \quad (25)$$

$$H_m(j) := \frac{A_{m, U_0}(j) + A_{m, T}(j)}{2}. \quad (26)$$

The earlier divisor-prefix analysis proves, uniformly on the packet,

$$|C_m(j)| + |H_m(j)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon. \quad (27)$$

As usual, every divisor-count and logarithmic loss below is absorbed by applying such an estimate with a smaller soft exponent.

The left and right kernel matrices are

$$K_{\gamma, \alpha}^L := \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_\alpha}(j) H_{m_\gamma}(j), \quad (28)$$

$$K_{\alpha, \gamma}^R := \sum_j \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) H_{m_\alpha}(j) C_{m_\gamma}(j). \quad (29)$$

Thus

$$B_\gamma^L = \sum_\alpha \gamma_\alpha^{(1)} K_{\gamma, \alpha}^L, \quad E_L = \sum_\gamma |B_\gamma^L|^2, \quad (30)$$

$$B_\alpha^R = \sum_\gamma \gamma_\gamma^{(2)} K_{\alpha, \gamma}^R, \quad E_R = \sum_\alpha |B_\alpha^R|^2. \quad (31)$$

The opposite outer coefficient is not included in E_L or E_R . TPC-33 proves the exact transfer

$$|\mathcal{S}_{\text{sh}}^{\text{all}}| \ll_{\mathcal{D}} \sqrt{Q \log(2L)} \left(\sqrt{E_L} + \sqrt{E_R} \right). \quad (32)$$

Consequently (1) implies

$$|\mathcal{S}_{\text{sh}}^{\text{all}}| \ll_{\varepsilon, h, \mathcal{D}} X^\varepsilon Q^2. \quad (33)$$

This paper improves the structural analysis of E_L, E_R ; it does not assume (1).

2.3 The row-gcd projector

TPC-33 proves the exact identity

$$\mathbf{1}_{(d,e) \leq G} = \sum_{\substack{v|d \\ v|e}} \lambda_G(v), \quad \lambda_G(v) = \sum_{\substack{g|v \\ g \leq G}} \mu(v/g). \quad (34)$$

For every squarefree d ,

$$\sum_{v|d} |\lambda_G(v)| \leq 3^{\omega(d)} \ll_{\varepsilon} d^{\varepsilon}. \quad (35)$$

This estimate is columnwise. We do not infer a small operator norm for the full mask from it.

2.4 The high- β packet

The endpoint specialization used in the ledger is

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}. \quad (36)$$

In particular,

$$\frac{T}{J} = X^{107/2000+o(1)} \longrightarrow \infty. \quad (37)$$

Whenever the shared-ultra argument is stated away from this packet, its explicit hypothesis is that the orbit-support diameter is $< T$. For the terminal-divisor splitting in [section 6](#), we also use the endpoint fact that, for sufficiently large X ,

$$T < \min_{\alpha,j} N_{\alpha}(j) \leq \max_{\alpha,j} N_{\alpha}(j) < U_0. \quad (38)$$

The first inequality follows from $T = X^{193/500+o(1)} = o(X)$ and $N_{\alpha}(j) \asymp_{\mathscr{D}} X$; the second is the chosen upper cutoff.

3 The exact physical Gram expansion

We first open the two structures that are invisible in the compact matrix K^L : the ultra divisor u and the row-gcd projector divisor v . All sums are finite, so the rearrangements below are identities.

3.1 A synthesis matrix for the left polarization

Let $\mathfrak{X}_L$ be the set of quadruples

$$\xi = (\alpha, j, u, v) \quad (39)$$

on the physical support such that

$$T < u \leq U_0, \quad u \mid N_{\alpha}(j), \quad v \mid d_{\alpha}. \quad (40)$$

Define

$$z_L(\xi) := \gamma_{\alpha}^{(1)} a_R(u) \lambda_G(v), \quad (41)$$

$$\Phi_{\gamma}^L(\xi) := \mathbf{1}_{v|d_{\gamma}} \mathfrak{B}_{\alpha,\gamma}(j) \mathbf{H}_{m_{\gamma}}(j). \quad (42)$$

The conditions in (40) are part of the index set and therefore are not repeated as indicators in z_L .

Theorem 3.1 (Exact opened TT^* identity). *Let $T_L : \mathbb{C}^{\mathfrak{X}_L} \rightarrow \mathbb{C}^{\{\gamma\}}$ be the synthesis matrix*

$$(T_L z)(\gamma) = \sum_{\xi \in \mathfrak{X}_L} \Phi_\gamma^L(\xi) z(\xi). \quad (43)$$

Then

$$B_\gamma^L = \sum_{\xi \in \mathfrak{X}_L} \Phi_\gamma^L(\xi) z_L(\xi), \quad (44)$$

$$\begin{aligned} E_L &= \|T_L z_L\|_2^2 \\ &= \sum_{\xi_1, \xi_2 \in \mathfrak{X}_L} \overline{z_L(\xi_1)} z_L(\xi_2) \mathcal{G}_L(\xi_1, \xi_2), \end{aligned} \quad (45)$$

$$\mathcal{G}_L(\xi_1, \xi_2) := \sum_{\gamma} \overline{\Phi_\gamma^L(\xi_1)} \Phi_\gamma^L(\xi_2) = (T_L^* T_L)(\xi_1, \xi_2). \quad (46)$$

In particular $\mathcal{G}_L$ is positive semidefinite. There is an identical right-polarized identity after exchanging α and γ .

Proof. Insert (25) and (34) into (30). The part of the physical multiplier not belonging to the row-gcd indicator is exactly $\mathfrak{B}_{\alpha, \gamma}(j)$. This gives (44). Squaring, summing over γ , and interchanging finite sums gives (45). The last identity is the definition of the adjoint product. $\square$

Remark 3.2 (Which diagonal is meant). The actual mask vanishes on the physical pair diagonal $\alpha = \gamma$. The new Gram expansion contains two independent copies α_1, α_2 of the *input* row. Its diagonal $\alpha_1 = \alpha_2$ is not deleted by the mask; each copy may still be different from the common opposite row γ . We call this the Gram-copy or same-input-row diagonal.

3.2 Divisor incidence in the Gram kernel

Proposition 3.3 (Least-common-multiple incidence). *Write $\xi_i = (\alpha_i, j_i, u_i, v_i)$. Uniformly on the physical packet,*

$$|\mathcal{G}_L(\xi_1, \xi_2)| \ll_{\varepsilon, \mathcal{D}} X^\varepsilon \left(L + \frac{Q}{[v_1, v_2]} \right). \quad (47)$$

The same estimate holds for the right Gram kernel.

Proof. A nonzero summand in (46) requires

$$v_1 \mid d_\gamma, \quad v_2 \mid d_\gamma,$$

and hence $[v_1, v_2] \mid d_\gamma$. For each permitted prime $\ell_\gamma \asymp L$, the number of integers $d_\gamma \asymp D$ divisible by $[v_1, v_2]$ is at most

$$O_{\mathcal{D}} \left(1 + \frac{D}{[v_1, v_2]} \right).$$

There are $O(L)$ possible prime labels, and the remaining factors in Φ_γ^L have an X^ε pointwise envelope by (27). Since $LD = Q$, this proves (47). The proof for the other polarization is the same. $\square$

Remark 3.4 (The principal projector layer). When $v_1 = v_2 = 1$, the right-hand side of (47) is $O(X^\varepsilon Q)$, the full opposite-row count. Thus the projector supplies a real gain on large-lcm layers but does not suppress its principal layer. The columnwise coefficient-mass bound (35) cannot change this conclusion.

3.3 The absolute baseline

Proposition 3.5 (Sign-blind baseline). *Without using cancellation between input rows or orbit points,*

$$\boxed{E_L + E_R \ll_{\varepsilon, \mathcal{D}} X^\varepsilon Q^3 J^2.} \quad (48)$$

Proof. By (27), (17), and the $O(J)$ orbit support,

$$|B_\gamma^L| \ll X^\varepsilon J \sum_\alpha |\gamma_\alpha^{(1)}| \ll X^\varepsilon Q J.$$

Summing its square over $O(Q)$ opposite rows proves the left estimate. The right estimate is symmetric. $\square$

The factor J^2 in (48) comes from coherent duplication of the two orbit times. The next section proves that several geometrically defined subfamilies are much smaller, but also shows where the unsigned incidence estimates proved here stop.

4 Unconditional Gram blocks and two-time orbit content

The full energy can be large even when each individual column entry has the expected pointwise size. We now identify the parts that are already small for purely structural reasons. Throughout this section a *mass* means the sum of absolute values of the specified terms in the opened Gram expansion; it is therefore an upper bound for the magnitude of that restricted contribution.

4.1 Closure of the Gram-copy diagonal

Expanding $E_L = \|K^L \gamma^{(1)}\|_2^2$ before opening divisors gives

$$E_L = E_{L,\Delta} + E_{L,\neq}, \quad (49)$$

$$E_{L,\Delta} := \sum_{\gamma, \alpha} |\gamma_\alpha^{(1)}|^2 |K_{\gamma, \alpha}^L|^2, \quad (50)$$

$$E_{L,\neq} := \sum_\gamma \sum_{\alpha_1 \neq \alpha_2} \gamma_{\alpha_1}^{(1)} \overline{\gamma_{\alpha_2}^{(1)}} K_{\gamma, \alpha_1}^L \overline{K_{\gamma, \alpha_2}^L}. \quad (51)$$

The off-diagonal term is real but need not be nonnegative. Define $E_{R,\Delta}, E_{R,\neq}$ symmetrically.

Theorem 4.1 (Unconditional Gram-copy diagonal closure). *On the physical packet,*

$$\boxed{E_{L,\Delta} + E_{R,\Delta} \ll_{\varepsilon, \mathcal{D}} X^\varepsilon Q^2 J^2.} \quad (52)$$

If $Q = X^{\beta+o(1)}$, $J = X^{1-\beta+o(1)}$, this is within the Q^3 budget whenever $\beta \geq 2/3$. At $\beta = 267/400$,

$$E_{L,\Delta} + E_{R,\Delta} \ll X^{o(1)} Q^3 X^{-1/400}. \quad (53)$$

Proof. The orbit and prefix envelopes give

$$|K_{\gamma, \alpha}^L| \ll X^\varepsilon J.$$

There are $O(Q)$ opposite rows. Hence

$$E_{L,\Delta} \ll X^\varepsilon Q J^2 \sum_\alpha |\gamma_\alpha^{(1)}|^2 \ll X^\varepsilon Q^2 J^2.$$

The right polarization is identical. Finally,

$$\frac{Q^2 J^2}{Q^3} = \frac{J^2}{Q} = X^{2-3\beta+o(1)}.$$

This is $O(X^{o(1)})$ for $\beta \geq 2/3$ and equals $X^{-1/400+o(1)}$ at $\beta = 267/400$. $\square$

Remark 4.2. This theorem concerns $\alpha_1 = \alpha_2$ in the duplicated Gram variables. It does not restore the deleted physical pair $\alpha = \gamma$; see [remark 3.2](#).

4.2 Exact target collisions

For a row-time pair define its opened absolute divisor mass

$$\mathfrak{d}(\alpha, j) := \left(\sum_{\substack{T < u \leq U_0 \\ u | N_\alpha(j)}} |a_R(u)| \right) \left(\sum_{v | d_\alpha} |\lambda_G(v)| \right). \quad (54)$$

The divisor bound and (35) imply

$$\mathfrak{d}(\alpha, j) \ll_\varepsilon X^\varepsilon. \quad (55)$$

Let $\mathcal{M}_{L,\text{tar}}$ be the absolute mass in (45) restricted by

$$m_{\alpha_1} j_1 = m_{\alpha_2} j_2. \quad (56)$$

This condition is equivalent to $N_{\alpha_1}(j_1) = N_{\alpha_2}(j_2)$.

Proposition 4.3 (Target-collision saving). *One has*

$$\boxed{\mathcal{M}_{L,\text{tar}} + \mathcal{M}_{R,\text{tar}} \ll_{\varepsilon, \mathcal{D}} X^\varepsilon Q^2 J.} \quad (57)$$

Proof. Put

$$r(p) = \#\{(\alpha, j) : m_\alpha j = p\}.$$

The row integer m_α determines α , and every representation supplies a divisor $m_\alpha \mid p$. Here $p \neq 0$ on the physical support because $m_\alpha \asymp Q$ and $j \asymp_{\mathcal{D}} J > 0$. Hence

$$r(p) \leq \tau(|p|) \ll_\varepsilon X^\varepsilon.$$

If

$$S(p) = \sum_{m_\alpha j = p} |\gamma_\alpha^{(1)}|,$$

then Cauchy–Schwarz gives

$$\sum_p S(p)^2 \leq \max_p r(p) \sum_p \sum_{m_\alpha j = p} |\gamma_\alpha^{(1)}|^2 \ll X^\varepsilon J \sum_\alpha |\gamma_\alpha^{(1)}|^2 \ll X^\varepsilon Q J.$$

The divisor masses in (55) are soft, and the crude Gram-kernel bound is $O(X^\varepsilon Q)$. Their product with the last display proves the left estimate. Exchange the row roles for the right estimate. $\square$

At the endpoint, $Q^2 J = Q^3 X^{-134/400 + o(1)}$. Thus this collision is not merely sparse relative to the absolute baseline; it is already well inside the required energy budget.

4.3 Shared ultra divisors

Let $\mathcal{M}_{L,u=}$ denote the absolute mass in the opened left Gram expansion with $u_1 = u_2$. The common value is automatically $> T$.

Proposition 4.4 (Shared-ultra incidence). *For an orbit interval of length $O_{\mathcal{D}}(J)$,*

$$\mathcal{M}_{L,u=} + \mathcal{M}_{R,u=} \ll_{\varepsilon, \mathcal{D}} X^\varepsilon Q^2 J \min \left\{ Q \left(1 + \frac{J}{T} \right), J \left(1 + \frac{Q}{T} \right) \right\}. \quad (58)$$

In particular, on the high- β packet,

$$\boxed{\mathcal{M}_{L,u=} + \mathcal{M}_{R,u=} \ll X^{o(1)} \frac{Q^3 J^2}{T}}. \quad (59)$$

If one also imposes $\alpha_1 = \alpha_2$ and the orbit-support diameter is $< T$, then

$$\mathcal{M}_{L,u=,\Delta} + \mathcal{M}_{R,u=,\Delta} \ll X^\varepsilon Q^2 J. \quad (60)$$

Proof. For the left polarization put

$$A_u = |a_R(u)| \sum_{\substack{\alpha, j \\ u | N_\alpha(j)}} |\gamma_\alpha^{(1)}|.$$

The projector-divisor masses are soft by (35). By (20), both $(u, m_\alpha) = 1$ and $(u, j) = 1$. For fixed α, u , the congruence

$$m_\alpha j \equiv -h \pmod{u}$$

selects one residue class modulo u . This gives

$$A_u \ll X^\varepsilon Q(1 + J/T).$$

For fixed j, u , the same congruence selects one row integer m_α modulo u . The row integers are distinct in an interval of length $O(Q)$, and the individual coefficients in (16) are $O(X^\varepsilon)$. Therefore

$$A_u \ll X^\varepsilon J(1 + Q/T).$$

Finally, each target has X^ε divisors, so

$$\sum_u A_u \ll X^\varepsilon QJ.$$

It follows that

$$\sum_u A_u^2 \leq (\max_u A_u) \sum_u A_u \ll X^\varepsilon QJ \min \{Q(1 + J/T), J(1 + Q/T)\}.$$

Multiplication by the $O(X^\varepsilon Q)$ opposite-row kernel proves (58).

At the endpoint, $T \gg J$ and $Q \gg T$, so the second member of the minimum is $X^{o(1)} JQ/T$; this gives (59).

For the same-input-row refinement, the two divisibilities imply

$$u \mid m_\alpha(j_1 - j_2).$$

Since $(u, m_\alpha) = 1$, $u \mid j_1 - j_2$. When the orbit diameter is $< T < u$, this forces $j_1 = j_2$. Summing $|\gamma_\alpha^{(1)}|^2$ over α , the $O(J)$ possible times, the soft divisor multiplicity, and $O(Q)$ opposite rows gives (60). The right side is symmetric. $\square$

Remark 4.5 (A saving, not closure). The estimate (59) saves the full factor $T = X^{193/500+o(1)}$ over (48). It nevertheless exceeds the Q^3 gate by

$$\frac{J^2}{T} = X^{279/1000+o(1)}.$$

Shared-divisor incidence is quantitatively useful here but does not close the layer.

4.4 The transposed target-content identity

TPC-30 studies two rows at one orbit time and uses common target content to constrain the row difference [7]. The Gram expansion naturally asks for the transpose: one row at two times.

Lemma 4.6 (Two-time target content). *Let $(m, h) = 1$. For all integers j, j' ,*

$$\boxed{(mj + h, mj' + h) = (mj + h, j' - j)}. \quad (61)$$

In particular, if $j \neq j'$, the common target content is at most $|j' - j|$.

Proof. The Euclidean algorithm gives

$$(mj + h, mj' + h) = (mj + h, m(j' - j)).$$

Since $(mj + h, m) = (h, m) = 1$, the factor m may be removed from the second argument. $\square$

This is an exact modular-return statement. Under the same hypothesis $(m, h) = 1$, for $c \geq 1$, a simultaneous hit at two times is equivalent to

$$c \mid N_m(j), \quad c \mid N_m(j') \iff (c, m) = 1, \quad c \mid j' - j, \quad j \equiv -hm^{-1} \pmod{c}, \quad (62)$$

where the inverse is defined by the displayed coprimality condition.

Proposition 4.7 (Large lag-content count). *Let $\mathcal{I}$ be an interval of J_0 consecutive integers and $(m, h) = 1$. For $1 \leq Z < J_0$,*

$$\begin{aligned} \#\{(j, j') \in \mathcal{I}^2 : j \neq j', (mj + h, mj' + h) > Z\} \\ \ll \frac{J_0^2}{Z} + J_0 \log(2J_0). \end{aligned} \quad (63)$$

Consequently the absolute left or right off-Gram mass restricted by large content on the common opposite row is

$$\ll_{\varepsilon, \mathcal{D}} X^\varepsilon Q^3 \left(\frac{J^2}{Z} + J \right). \quad (64)$$

Proof. For $0 < |r| < J_0$, every pair with $j' - j = r$ and $\gcd > Z$ contributes to

$$\sum_{\substack{c \mid r \\ c > Z \\ (c, m) = 1}} \mathbf{1}_{j \equiv -hm^{-1} \pmod{c}}.$$

For fixed r, c , the number of possible j 's is at most $J_0/c + 1$. Summing first over multiples of c gives

$$\ll J_0^2 \sum_{c > Z} \frac{1}{c^2} + J_0 \sum_{Z < c < J_0} \frac{1}{c} \ll \frac{J_0^2}{Z} + J_0 \log(2J_0).$$

For (64), sum this count over $O(Q)$ common opposite rows and use $(\sum_\alpha |\gamma_\alpha^{(i)}|)^2 \ll Q^2$, together with all pointwise envelopes. $\square$

The $j = j'$ temporal diagonal has content $N_m(j) \asymp X$ and absolute scale $Q^3 J$. For $j \neq j'$, choosing $Z = J^{1-\eta}$ in (64) saves almost one orbit factor over the $Q^3 J^2$ baseline, but the remaining $J^{1+\eta}$ scale still does not cross the Q^3 gate. The result is therefore a rigorous pruning theorem, not a substitute for signed off-diagonal cancellation.

5 Orbit-sliced row energy

The preceding TT^* expansion duplicates both row and time. The matched polarization has an additional structure: on the left, the half-prefix depends on (γ, j) but not on α . We can therefore compress the two-time problem to a same-time row energy at the cost of one explicit factor J .

5.1 Definition and exact transfer

Define

$$Y_{\gamma,j}^L := \sum_{\alpha} \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad (65)$$

$$V_L := \sum_{\gamma,j} |Y_{\gamma,j}^L|^2. \quad (66)$$

For the right polarization put

$$Y_{\alpha,j}^R := \sum_{\gamma} \gamma_{\gamma}^{(2)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_{\gamma}}(j), \quad (67)$$

$$V_R := \sum_{\alpha,j} |Y_{\alpha,j}^R|^2. \quad (68)$$

No separation of the joint multiplier is used in these definitions.

Theorem 5.1 (Orbit-sliced energy transfer). *On the physical packet,*

$$\boxed{E_L + E_R \ll_{\varepsilon, \mathscr{D}} X^{\varepsilon} J (V_L + V_R).} \quad (69)$$

Consequently the additional estimate

$$\boxed{V_L + V_R \ll_{\varepsilon, h, \mathscr{D}} X^{\varepsilon} \frac{Q^3}{J}} \quad (70)$$

implies the inherited Q^3 energy gate (1), and hence the scalar shell bound (33).

Proof. The definitions give the exact factorization

$$B_{\gamma}^L = \sum_j H_{m_{\gamma}}(j) Y_{\gamma,j}^L.$$

By Cauchy–Schwarz, the $O(J)$ support, and (27),

$$|B_{\gamma}^L|^2 \ll X^{\varepsilon} J \sum_j |Y_{\gamma,j}^L|^2.$$

Summing over γ gives $E_L \ll X^{\varepsilon} J V_L$. The right factorization

$$B_{\alpha}^R = \sum_j H_{m_{\alpha}}(j) Y_{\alpha,j}^R$$

is identical. This proves (69); the conditional implications follow from (32). $\square$

Remark 5.2 (Where the factor J is spent). The Cauchy step in the proof is sharp for a time-constant coherent sequence. Thus (69) does not itself create cancellation. Its value is that it trades the original two-time Gram kernel for a single-time row correlation with a completely explicit loss.

5.2 The natural diagonal scale

Expanding V_L in the two copies of the input row, define

$$V_{L,\Delta} := \sum_{\gamma,j,\alpha} |\gamma_\alpha^{(1)}|^2 |\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)|^2 |C_{m_\alpha}(j)|^2, \quad (71)$$

$$V_{L,\neq} := V_L - V_{L,\Delta}, \quad (72)$$

and similarly on the right.

Proposition 5.3 (Orbit-sliced diagonal). *One has*

$$\boxed{V_{L,\Delta} + V_{R,\Delta} \ll_{\varepsilon,\mathscr{D}} X^\varepsilon Q^2 J.} \quad (73)$$

At the high- β endpoint,

$$\frac{Q^3/J}{Q^2 J} = \frac{Q}{J^2} = X^{1/400+o(1)}. \quad (74)$$

Proof. For each α , there are $O(QJ)$ pairs (γ, j) , while the multiplier and increment have an X^ε pointwise envelope. Hence

$$V_{L,\Delta} \ll X^\varepsilon QJ \sum_{\alpha} |\gamma_\alpha^{(1)}|^2 \ll X^\varepsilon Q^2 J.$$

The other polarization is symmetric. The ratio in (74) is immediate from (36). $\square$

The allowed energy (70) is therefore only $X^{1/400+o(1)}$ above the natural diagonal size. In particular, the stronger estimate

$$\boxed{V_L + V_R \ll X^\varepsilon Q^2 J} \quad (75)$$

would imply

$$E_L + E_R \ll X^\varepsilon Q^2 J^2 \ll_{\mathscr{D}} X^{2+\varepsilon} = X^{\varepsilon+o(1)} Q^3 X^{-1/400}. \quad (76)$$

Estimate (75) is not proved here.

5.3 A uniform off-diagonal operator gate

For each j , let M_j^L be the matrix

$$(M_j^L)_{\gamma,\alpha} = \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_\alpha}(j). \quad (77)$$

Then

$$V_L = \left\langle \left(\sum_j (M_j^L)^* M_j^L \right) \gamma^{(1)}, \gamma^{(1)} \right\rangle. \quad (78)$$

Write this positive Gram matrix as

$$G_L^{\text{orb}} = D_L^{\text{orb}} + O_L^{\text{orb}}, \quad (79)$$

where D_L^{orb} is its entrywise diagonal and O_L^{orb} its Hermitian off-diagonal part. Define the right objects symmetrically.

Proposition 5.4 (Operator-norm sufficient condition). *The uniform bounds*

$$\boxed{\|O_L^{\text{orb}}\|_{\text{op}} + \|O_R^{\text{orb}}\|_{\text{op}} \ll_{\varepsilon,h,\mathscr{D}} X^\varepsilon \frac{Q^2}{J}} \quad (80)$$

imply (70) whenever $J^2 \ll X^\varepsilon Q$.

Proof. The diagonal entries of G_L^{orb} are

$$\sum_{\gamma, j} |\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j)|^2 |C_{m_\alpha}(j)|^2 \ll X^\varepsilon QJ.$$

Thus $\|D_L^{\text{orb}}\|_{\text{op}} \ll X^\varepsilon QJ$, and similarly on the right. Combine (18) with (80):

$$V_L + V_R \ll X^\varepsilon \left(QJ + \frac{Q^2}{J} \right) Q.$$

If $J^2 \ll X^\varepsilon Q$, both terms are $O(X^\varepsilon Q^3/J)$. $\square$

Remark 5.5. The operator condition is stronger than the coefficient-specific estimate (70); it is a convenient uniform sufficient condition, not an equivalent reformulation. A Hilbert–Schmidt second moment or a Schur row-sum estimate for $O_L^{\text{orb}}, O_R^{\text{orb}}$ would each be sufficient if it reached the displayed scale.

The dynamical meaning is modest but exact. The two copies j, j' in the original Gram expansion are two times on the affine orbit. Orbit slicing removes their coherent summation at a known cost and leaves the same-time row autocorrelation. No mixing, spectral gap, or Möbius decorrelation has been inferred.

6 The terminal Möbius layer and a sign-blind obstruction

The orbit-sliced gate is a same-time row energy, and the unsigned incidence estimates developed here do not by themselves control it. On the high- β endpoint packet this becomes explicit before any affine reflection: the largest divisor of every target lies in the ultra-divisor range, with a coefficient that may vanish when the target is not squarefree.

6.1 Exact terminal/proper-divisor splitting

On the endpoint packet, (38) gives $T < N_m(j) < U_0$. Therefore

$$C_m(j) = C_m^{\text{term}}(j) + C_m^{\text{prop}}(j), \tag{81}$$

$$C_m^{\text{term}}(j) := a_R(N_m(j)) = -\mu(N_m(j)) \log N_m(j), \tag{82}$$

$$C_m^{\text{prop}}(j) := \sum_{\substack{T < u < N_m(j) \\ u|N_m(j)}} a_R(u). \tag{83}$$

This is an exact identity, including when $\mu(N_m(j)) = 0$.

Insert (81) into (65) and write

$$Y_{\gamma, j}^L = Y_{\gamma, j}^{L, \text{term}} + Y_{\gamma, j}^{L, \text{prop}}, \tag{84}$$

with corresponding squared energies V_L^{term} and V_L^{prop} ; define the right versions similarly. One always has

$$V_L \leq 2(V_L^{\text{term}} + V_L^{\text{prop}}), \tag{85}$$

and similarly on the right. This inequality gives a sufficient layerwise route, but no converse lower bound: terminal and proper-divisor vectors may cancel before squaring.

6.2 The fused terminal sign

Let $m = \ell d$. The row coefficient (16) contains the sign $\mu(d)$, while (82) contains $-\mu(\ell dj + h)$. Target primitivity gives

$$(d, \ell dj + h) = 1. \quad (86)$$

Hence, whenever the two Möbius factors are nonzero,

$$\mu(d)\mu(\ell dj + h) = \mu(d(\ell dj + h)). \quad (87)$$

On the principal projector layer $v = 1$, the terminal divisor has complementary factor $s := N_m(j)/u = 1$. Freezing (ℓ, j) , the resulting row sum has the schematic but literal sign pattern

$$\sum_{d \lesssim D} \mu(d)\mu(\ell jd + h)W_{\ell, j, \gamma}(d), \quad (88)$$

where $W_{\ell, j, \gamma}$ retains the actual smooth, residue, and opposite-row mask factors. After restoring the ℓ -sum and expanding its same-time square, one obtains

$$\mu(d_1)\mu(\ell_1 d_1 j + h)\mu(d_2)\mu(\ell_2 d_2 j + h). \quad (89)$$

The averaging over $j, \ell_1, \ell_2, d_1, d_2$ and the common opposite row makes this weaker than a pointwise theorem for every affine direction. It is nevertheless sign-sensitive: the congruence and gcd identities proved above do not remove (89).

Remark 6.1 (Relation to Chowla technology). TPC-33 observes that the unrestricted fixed-direction family contains $\sum_n \mu(n)\mu(n+1)$, whereas logarithmically averaged fixed-form results and shift-averaged results have different hypotheses [4, 5, 10]. We do not claim that the physically averaged operator gate (80) is equivalent to full two-point Chowla. We claim only that its terminal layer is a genuine averaged affine Möbius correlation and is not controlled by the unsigned incidence estimates in this paper.

6.3 A sharp abstract majorant model

The next finite example shows why closing the Gram-copy diagonal does not close the full Gram energy. It also records the precise limitation of a sign-blind terminal majorant.

Proposition 6.2 (Coherent zero-pair-diagonal model). *Let the input and opposite row sets both have $M \geq 2$ elements, let the orbit have $J_0 \geq 1$ times, and take*

$$\gamma_\alpha = 1, \quad \mathbf{H}_\gamma(j) = 1, \quad C_\alpha(j) = 1, \quad \mathfrak{A}_{\alpha, \gamma}(j) = \mathbf{1}_{\alpha \neq \gamma}. \quad (90)$$

Then

$$K_{\gamma, \alpha} = J_0 \mathbf{1}_{\alpha \neq \gamma}, \quad (91)$$

$$E = M(M-1)^2 J_0^2, \quad (92)$$

$$E_\Delta = M(M-1)J_0^2, \quad (93)$$

$$E_{\neq} = M(M-1)(M-2)J_0^2. \quad (94)$$

The orbit-sliced energy is

$$V = M(M-1)^2 J_0, \quad E = J_0 V. \quad (95)$$

Proof. For every opposite row γ , exactly $M-1$ input rows survive. Thus $B_\gamma = (M-1)J_0$, which proves (92). There are $M(M-1)$ nonzero matrix entries, each of squared modulus J_0^2 , proving (93); subtraction gives the off-diagonal formula. At every (γ, j) , the orbit-sliced row sum is $M-1$, proving (95). $\square$

For $M \asymp Q$ and $J_0 \asymp J$, this model has

$$E \asymp Q^3 J^2, \quad V \asymp Q^3 J.$$

It respects the equal-pair deletion, the row cardinality, both row norm scales, the orbit length, and all pointwise envelopes. It may also be viewed as retaining one abstract terminal atom per row-time pair and replacing its fused sign by $+1$. It is *not* asserted to be the physical divisor kernel.

Corollary 6.3 (Envelope-only method boundary). *No estimate that uses only the listed cardinalities, pointwise envelopes, and equal-pair deletion can uniformly improve (48) to $O(Q^3)$ over the corresponding abstract majorant class.*

The corollary does not say that the physical energy is large and does not exclude a stronger theorem using additional arithmetic incidence. The specific unsigned incidence bounds proved here save factors on selected collision layers but remain above the complete gate. A physical proof may use the fused signs in (87), cancellation between terminal and proper-divisor layers in (84), or another sign-sensitive spectral mechanism. The listed sign-blind data alone do not supply the missing powers. This is a local method obstruction consistent with the classical parity boundary [2]; it is not a parity-breaking theorem.

7 The high- β ledger and the next gate

The preceding estimates have different meanings: some cross the Q^3 threshold, some merely save a full structural factor, and some are conditional reductions. We collect them with a common exponent denominator before choosing the next route.

7.1 Exact endpoint comparison

At

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)},$$

the exponent ledger is as follows.

Table 1: Physical energy and orbit-sliced scales at the endpoint.

Object	Bound or target	X -exponent
Required column energy	Q^3	4005/2000
Absolute two-time baseline	$Q^3 J^2$	5335/2000
Gram-copy diagonal	$Q^2 J^2$	4000/2000
Exact target collision	$Q^2 J$	3335/2000
Shared ultra divisor	$Q^3 J^2 / T$	4563/2000
Absolute orbit-sliced baseline	$Q^3 J$	4670/2000
Sufficient orbit-sliced gate	Q^3 / J	3340/2000
Orbit-sliced diagonal	$Q^2 J$	3335/2000
Diagonal-scale orbit target	$Q^2 J$	3335/2000

The table has four immediate consequences.

- (i) The Gram-copy diagonal closes with margin $5/2000 = 1/400$.
- (ii) The exact target collision lies $670/2000 = 134/400$ below the energy target.
- (iii) Shared-ultra incidence saves the entire factor T from the absolute baseline, but still misses the energy gate by $558/2000 = 279/1000$, which is J^2/T .

- (iv) The orbit-sliced diagonal lies only $5/2000 = 1/400$ below the allowed orbit energy. Thus the new interface is quantitatively tight: it asks for a near-diagonal-scale row theorem, not a generic square-root estimate on each original column.

7.2 The remaining coefficient-specific form

Write

$$V_L + V_R = V_{L,\Delta} + V_{R,\Delta} + V_{L,\neq} + V_{R,\neq}. \quad (96)$$

The first two terms are controlled by [proposition 5.3](#). A direct coefficient-specific sufficient condition is therefore

$$|V_{L,\neq}| + |V_{R,\neq}| \ll X^\varepsilon \frac{Q^3}{J}. \quad (97)$$

The off-diagonal quantities are real but may have either sign; replacing them by their termwise absolute values would return to the failed sign-blind scale.

The uniform operator gate (80) is stronger than (97). Two standard sufficient interfaces for it are

$$\sum_{\alpha_1 \neq \alpha_2} |O_L^{\text{orb}}(\alpha_1, \alpha_2)|^2 \ll X^\varepsilon \frac{Q^4}{J^2}, \quad (98)$$

$$\max_{\alpha_1} \sum_{\alpha_2 \neq \alpha_1} |O_L^{\text{orb}}(\alpha_1, \alpha_2)| \ll X^\varepsilon \frac{Q^2}{J}, \quad (99)$$

with the analogous right-hand estimates. The first is a Hilbert–Schmidt second moment and the second a Schur row-sum bound. Neither is proved here, and neither should be confused with a consequence of the diagonal estimate.

7.3 A route menu

The exact reductions leave several nonexclusive ways forward.

Route A: *Same-time projective energy*. Apply the finite fixed-determinant Plancherel geometry of TPC-33 after orbit slicing, using several moduli or an integer large sieve so that the physical slope set is averaged without the single-modulus no-wrap conflict. The target is (98), not pointwise cancellation on every affine line.

Route B: *Terminal/proper dispersion*. Use (81). The proper-divisor layers retain a nontrivial complementary factor and may admit bilinear dispersion. The terminal layer requires control of the fused-sign correlation $\mu(d)\mu(\ell dj + h)$, cancellation between this layer and the proper layers, or stronger arithmetic incidence not captured by the estimates proved here.

Route C: *Lag-content plus zero-lag analysis*. Combine [proposition 4.7](#) with a dyadic decomposition in the modular-return divisor. Large lag content is sparse; the zero-lag and small-content pieces must then be treated by a signed row-operator estimate or by additional arithmetic incidence beyond the bounds proved here.

Route D: *Restricted averaged Chowla input*. Prove only the physically weighted family average in (97). This is weaker than a uniform theorem for every growing affine direction, but stronger than the sign-blind incidence currently available.

The first route is the most compatible with the average projective identity already proved; the second exposes most clearly where parity-sensitive signs enter the present interface. A failed attempt along either route remains informative if it produces a sharp operator obstruction or a quantitative range in which one of the proper-divisor layers closes.

7.4 Relation to the preceding chain

TPC-28 closes selected truncated base packets before the present ultra-long shell [6]. TPC-30 supplies the two-row, one-time target-content geometry [7]. TPC-31 develops determinant cells and proposes a matched TT^* route [8]; it does not perform the Gram-copy closure above. TPC-32 supplies the matched polarization and the distinguished zero-frequency gate [9]. TPC-33 defines the physical column energies and proves their scalar transfer [10]. The new step is the exact Gram expansion of those energies, unconditional closure of its copy diagonal and target collision, the transposed two-time content theorem, and the orbit-sliced reduction.

Modulo a divisor c , the map $j \mapsto mj + h$ is an affine traversal with invertible step. A simultaneous hit of the zero class at two times is precisely the return condition $c \mid j' - j$. This is deterministic modular recurrence geometry. It is not a claim of mixing, spectral decay, a logistic-orbit correspondence, or Möbius disjointness.

Even a future proof of (70) would close only the inherited matched residual packet. Later reconstruction, main-term, and positivity stages would remain. None of the results above implies a Hardy–Littlewood asymptotic or infinitely many twin primes.

8 Exact certificate and claim boundary

The accompanying standard-library certificate is

`experiments/tpc34_certificate.py.`

It uses integers, `Fraction`, and a hand-written Gaussian-rational arithmetic. It contains no floating-point calculation, random input, or Python `assert` statement. Running

```
python experiments/tpc34_certificate.py
python -O experiments/tpc34_certificate.py
```

performs the same checks and regenerates byte-identical JSON.

8.1 Finite checks

The archived run performs 213,976 exact checks.

Table 2: Exact finite checks in the TPC-34 certificate.

Check family	Count
Complex TT^* , Gram split, and slice identities	106
Opened gcd-projector and lcm incidence	3,368
Weighted target-collision fibers	255
Shared-ultra congruence and occupancy	2,379
Two-time gcd, modular returns, and lag counts	172,434
Orbit-sliced Cauchy and sharpness	80
Coherent zero-pair-diagonal model	35,340
High- β rational ledger	14
Total	213,976

Gaussian-rational tests verify the standard Gram convention

$$\mathcal{G}(\xi_1, \xi_2) = \sum_{\gamma} \overline{\Phi_{\gamma}(\xi_1)} \Phi_{\gamma}(\xi_2), \quad E = \sum_{\xi_1, \xi_2} \overline{z(\xi_1)} z(\xi_2) \mathcal{G}(\xi_1, \xi_2). \quad (100)$$

This detects a conjugate or transpose error that real-valued examples would miss. The opened-projector tests compare the direct masked column with the (u, v) -expanded column and then compare their energies. Since $\log u$ is not rational, the test specializes the arbitrary real ultra coefficient to

$$a_{\text{test}}(u) = -\mu(u)(u + 1).$$

This certifies the finite linear identity for one nontrivial exact coefficient system; the proof in the text establishes it for the actual logarithmic coefficients.

The collision suite verifies

$$\#\{m_1 j_1 = m_2 j_2\} = \sum_p r(p)^2, \quad r(p) \leq \tau(p),$$

including the weighted Cauchy inequality used in [proposition 4.3](#). The shared-ultra suite checks both fixed-row and fixed-time residue occupancy. The two-time suite verifies (61), the modular-return equivalence (62), and an explicit harmonic upper bound for the divisor overcount in [proposition 4.7](#). The coherent model is checked entry by entry rather than only through its closed formula.

8.2 Reproducibility values

The archived outputs have:

$$\begin{array}{ll} \text{normalized source SHA-256} & 92\text{db9e309593faa815b1806cc4a656bf} \\ & \text{c02b4ffab5eb1eb4f61c7b973a24168f}, \end{array} \quad (101)$$

$$\begin{array}{ll} \text{JSON SHA-256} & 50\text{e7d0aefc9dd74f549831c8f70072fb} \\ & 34\text{be4788ae9d193e7acf79753d8c34}, \end{array} \quad (102)$$

$$\begin{array}{ll} \text{certificate digest} & \text{da2a05161550265582a881a0b7017a3b} \\ & 0\text{be540c62df3cab187c58dbeaab9d9be}. \end{array} \quad (103)$$

Line endings are normalized before the source hash is taken. The certificate digest hashes the canonical JSON payload before its digest field is appended.

8.3 What is not certified

Finite enumeration verifies algebra, exact finite incidence, and rational bookkeeping. It does not prove:

- (i) the asymptotic divisor or prefix X^ε envelopes;
- (ii) the growing-parameter shared-ultra estimate;
- (iii) the off-diagonal operator bound (80);
- (iv) the orbit-sliced energy gate (70);
- (v) the inherited Q^3 column-energy gate;
- (vi) cancellation in an affine Möbius or Chowla family;
- (vii) the distinguished zero-frequency flatness inequality;
- (viii) closure of every earlier residual packet;
- (ix) positivity or a Hardy–Littlewood main term;
- (x) infinitely many twin primes; or

(xi) a breach of sieve parity.

The unconditional analytic results of this paper are the exact physical Gram expansion, the lcm kernel bound, Gram-copy diagonal closure, target collision estimate, shared-ultra pruning, two-time target-content and lag count, orbit-sliced transfer and diagonal estimate, and the precisely scoped envelope-only obstruction. The next unproved analytic input is (70), or one of its sufficient operator forms.

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